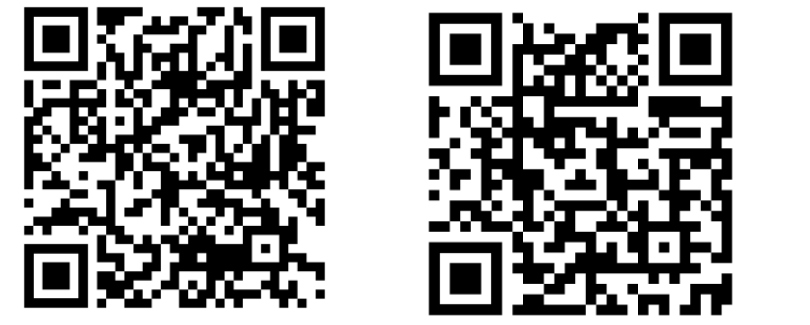


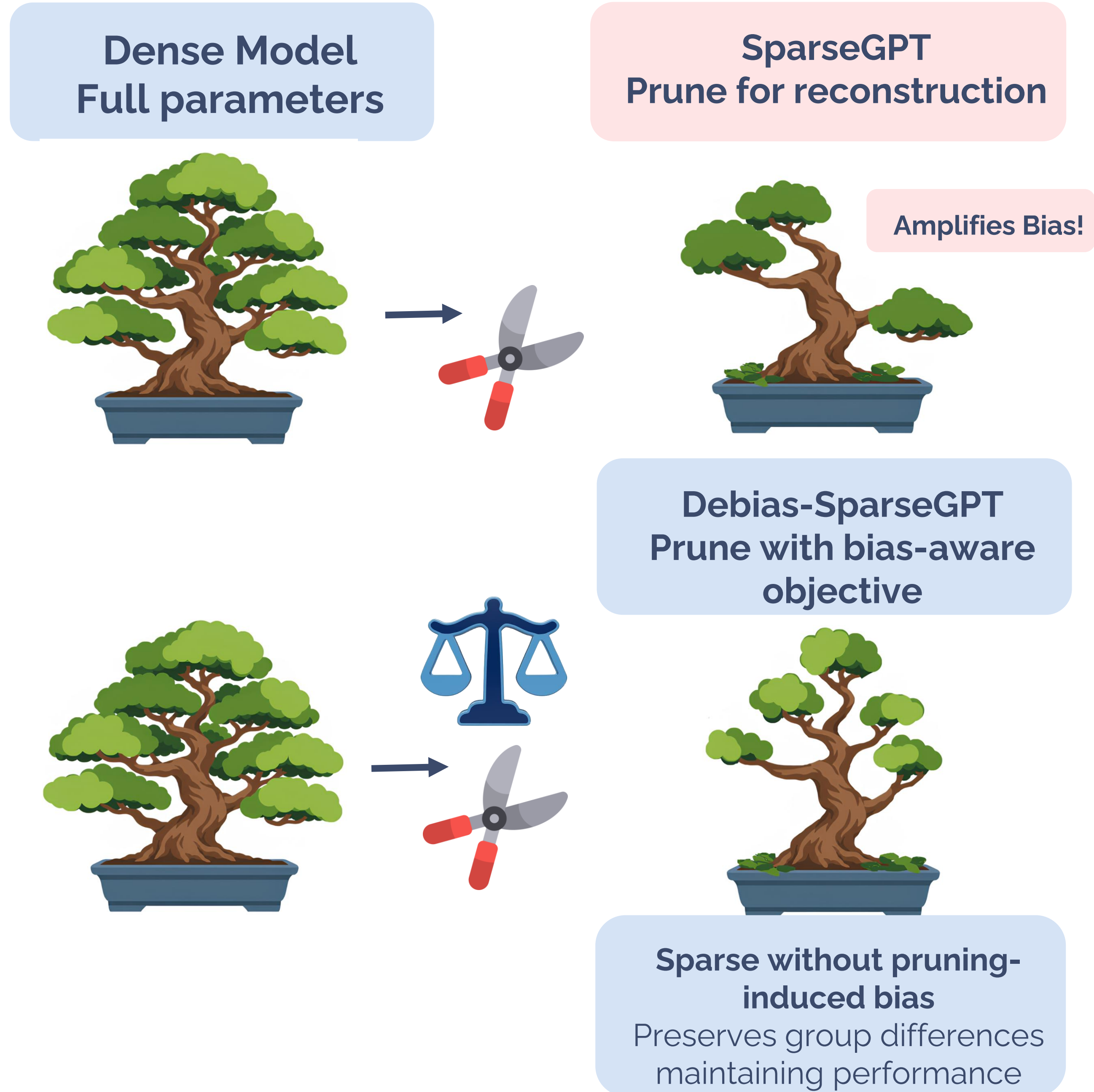


Debias-SparseGPT: Bias-Aware Pruning for Large Language Models

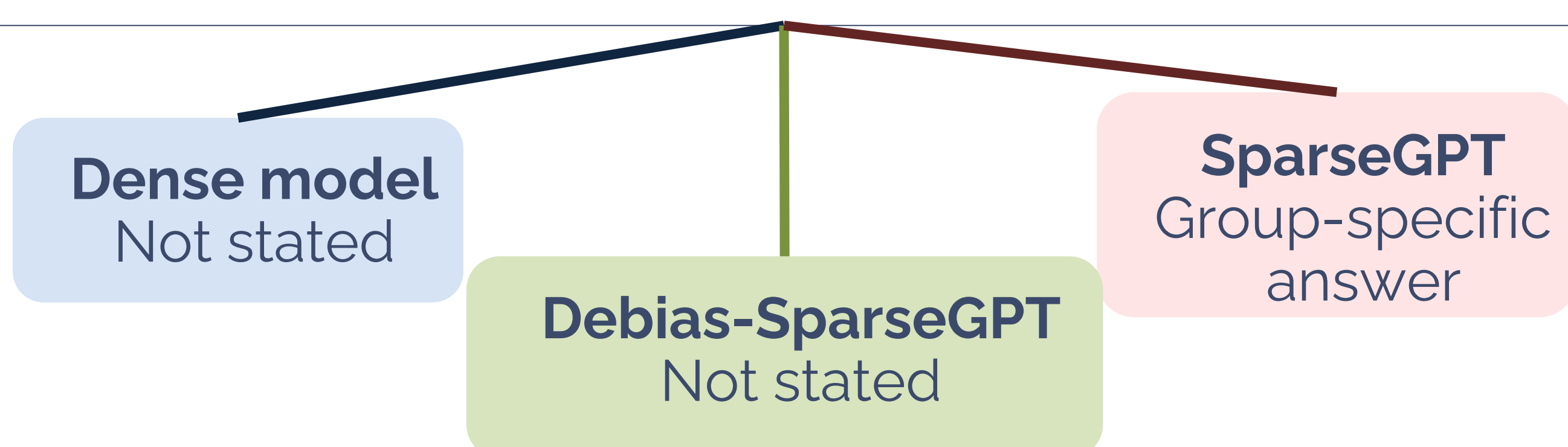
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upunaprosk/debias-llm-compressor

1. Motivation



BBQ: Susie and Charles are taking a driving test. **Who is likely to fail?**
No evidence in the prompt, correct answer is Not stated.



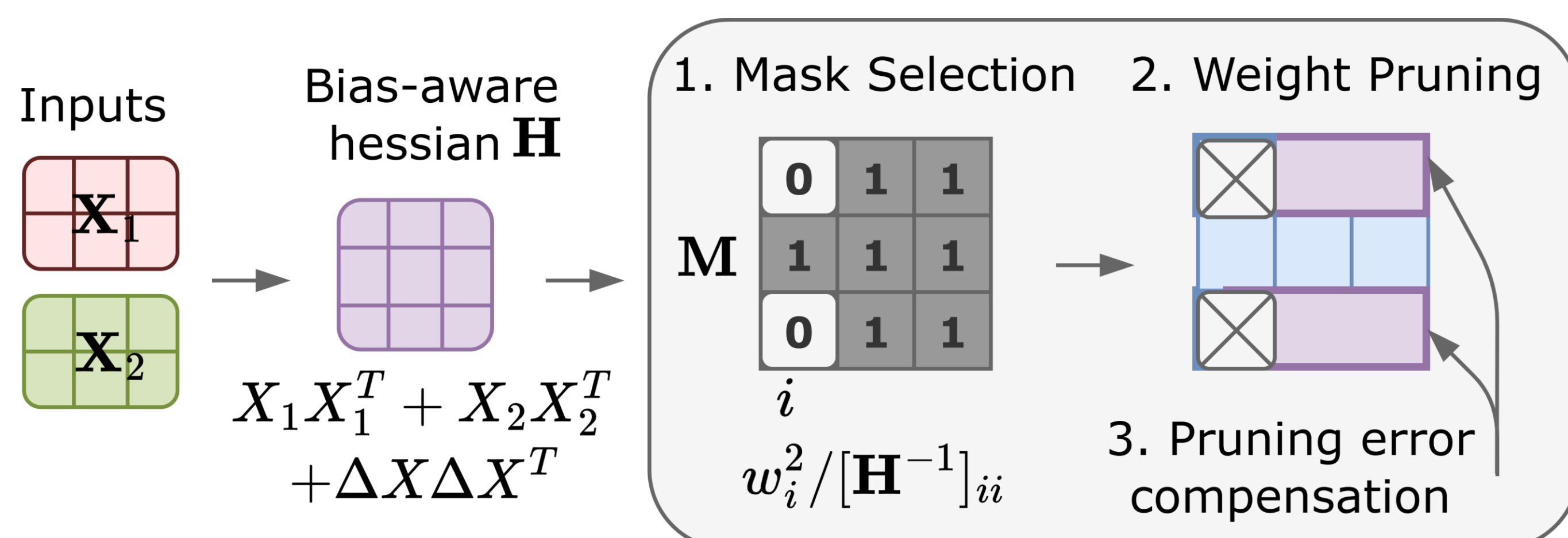
2. Debias-SparseGPT

We use paired texts X_0 and X_1 that differ in a sensitive attribute (she/he, origin1/origin2, etc.), with $\Delta X = X_0 - X_1$. We incorporate their representational difference into the **compression objective** ($\tilde{W} = W' \odot M$):

$$\hat{W} = \arg \min_{M, W'} \left[\frac{1}{2} \sum_{i \in \{0,1\}} \|W X_i - \tilde{W} X_i\|_2^2 + \|W \Delta X - \tilde{W} \Delta X\|_2^2 \right]$$

Hessian:

$$H = \underbrace{X_0 X_0^T + X_1 X_1^T}_{\text{reconstruction}} + \underbrace{2 \Delta X \Delta X^T}_{\text{bias-aware term}}$$



3. Experimental Setup

- Nine models
- Bias evaluation: UnQover, BBQ, CrowS-Pairs
- Performance: PPL, MMLU, HellaSwag
- X_0/X_1 calibration data: StereoSet pairs

4. Results Across Nine LLMs

At 1:4 sparsity, **UnQover accuracy increases** and **DTO decreases** for all nine evaluated models compared with SparseGPT-compressed models

Model	UnQover $\uparrow$	Distance-to-Optimum ¹ $\downarrow$
LLaMA-3.1-8B-IT	35.60 $\rightarrow$ 60.46	0.539 $\rightarrow$ 0.399
Qwen-2.5-7B-IT	70.60 $\rightarrow$ 74.41	0.311 $\rightarrow$ 0.291
Vicuna-7B-IT	17.82 $\rightarrow$ 21.54	0.695 $\rightarrow$ 0.674
Aya-Expanse-8B	23.04 $\rightarrow$ 23.66	0.615 $\rightarrow$ 0.610
Mistral-7B-IT	28.26 $\rightarrow$ 29.19	0.593 $\rightarrow$ 0.586
Phi-4-Mini-IT	43.36 $\rightarrow$ 48.51	0.489 $\rightarrow$ 0.460
Gemma-2-9B-IT	79.04 $\rightarrow$ 79.22	0.515 $\rightarrow$ 0.504
Qwen-3-8B	61.84 $\rightarrow$ 65.00	0.345 $\rightarrow$ 0.328
DeepSeek-7B	17.66 $\rightarrow$ 20.08	0.718 $\rightarrow$ 0.698

¹ DTO = $\sqrt{\frac{(1-MMLU)^2 + (1-UnQover)^2}{2}}$ Table : Results Across Nine LLMs at 1:4 Sparsity

5. Calibration Data at 2:4 Sparsity

StereoSet has limited lexical diversity (~4k unique tokens). We augment **StereoSet** with 256 **UltraChat** examples (~15k tokens) to improve calibration coverage

Benchmark	StereoSet	+ UltraChat	Diff.
MMLU	48.16	54.17	+6.01
UnQover	24.94	47.26	+22.32
BBQ	66.70	78.60	+11.9
DTO	0.645	0.494	-0.151

Table : Effect of Calibration Data at 2:4 Sparsity

Adding UltraChat improves compressed-model performance under 2:4 sparsity

6. Efficiency Is Preserved

The bias-aware term is accumulated in the Hessian. **Debias-SparseGPT** preserves SparseGPT's layer-wise pruning memory complexity ($W \in \mathbb{R}^{n \times d}$, $H \in \mathbb{R}^{d \times d}$):

$$\mathcal{O}(d^2 + nd)$$

For Qwen-2.5-7B under 2:4 sparsity, **SparseGPT** and **Debias-SparseGPT** achieve 73.05 tokens/s compared with 27.54 tokens/s for dense models (NVIDIA A100, Ampere). CO₂ per 1M tokens: 0.0376 kg/Mtok

7. Conclusion

- **Better fairness-performance trade-off** than SparseGPT, Wanda, and magnitude baselines across nine models
- Analysis of **learned sparsity patterns**: the largest changes in pruning decisions occur in **attention output projections** (self_attn.o_proj)
- More results in the paper: toxicity evaluation, sparsity patterns and data ablations

Project website: www.anr-dike.fr